Name _____
Practice Derivatives

Practice with Derivatives: No Calculator Allowed

Part I: Use your knowledge of derivatives to find $\frac{dy}{dx}$ for each of the following:

1. $y = 3x^3 - 4x^2 + 2x - 5$

2. $y = \frac{1}{x}$

3. $y = x\sqrt{x}$

4. Write an equation of the line tangent to the graph of $y = \sqrt[3]{x}$ at $x = 8$.

5. Write an equation of the line tangent to the graph of $y = 3 - \frac{4}{x^2}$ at $x = 2$.

6. Write an equation of the normal line to the graph of $f(x) = 3\cos x$ at the point $\left(\frac{\pi}{2}, 0\right)$.

Part II: Show <u>all</u> work then write your choice in the space provided.

7. If $y=\frac{1-x}{x-1}$, then $\frac{dy}{dx}=$

(A) - 1 (B) 0 (C) $\frac{-1}{x-1}$ (D) $\frac{-2}{x-1}$ (E) $\frac{-2x}{(x-1)^2}$

7)_____

---

8. What is the 2004th derivative of $\sin x$?

(A) $-\cos x$ (B) $\cos x$ (C) $\sin x$ (D) $-\sin x$ (E) 0

8)_____

---

9. The equation of the line tangent to the curve $y=\frac{kx+8}{k+x}$ at $x=-2$ is $y=x+4$. What is the value of k?

(A) - 3 (B) – 1 (C) 1 (D) 3 (E) 4

9)_____